

AN EXTERIOR-POINT PROOF OF TWO COMPLETED-INTERLACING CONJECTURES

ABSTRACT. We prove Conjectures 1 and 2 of Jordaan and Kumar on two-point completed interlacing for Meixner–Pollaczek and pseudo-Jacobi polynomials. The common mechanism is an exterior-point criterion: in the relevant mixed recurrence, the zero of the linear coefficient lies outside the interval between the two completion points. For pseudo-Jacobi polynomials, we also give a three-term-recurrence proof, independent of orthogonality, of the consecutive-degree interlacing needed throughout the conjectured parameter range.

1. INTRODUCTION

For monic polynomials G and Q of degrees $n + 1$ and n , respectively, we write $G \prec Q$ when their zeros are real and simple and, in increasing order, satisfy

$$y_1 < q_1 < y_2 < \cdots < q_n < y_{n+1}.$$

Jordaan and Kumar developed a general framework for completing interlacing by two additional points [2]. In the configuration relevant here, their Theorem 2.2(e) yields the desired completed interlacing provided that, inside the gap of the higher-degree polynomial containing a completion point, two consecutive zeros of the lower-degree polynomial straddle that point: with E_1, E_2 the two completion points, $y_{1,n+1} < \cdots < y_{n+1,n+1}$ the zeros of the higher-degree polynomial and $z_{1,n} < \cdots < z_{n,n}$ those of the lower-degree one, the hypothesis is

$$y_{k,n+1} < z_{l,n} < E_1 < z_{l+1,n} < y_{k+1,n+1}$$

for the first of the two conclusions, and the analogous condition at E_2 for the second. Their Conjectures 1 and 2 assert, for Meixner–Pollaczek and pseudo-Jacobi polynomials, that the corresponding completed interlacing holds without this extra placement hypothesis whenever the two completion points lie in distinct interior gaps of the higher-degree polynomial.

We prove both conjectures. The key observation is that each mixed recurrence has the form

$$A(x)P(x) = -(x - c)G(x) + (x - r_-)(x - r_+)Q(x),$$

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and the distinguished point c lies outside $[r_-, r_+]$. A direct sign count then forces the required local zero configuration and proves the completed interlacing without assuming in advance that P is real-rooted or coprime to G . The two conjectures are vacuous for $n = 1$, so the applications below are stated for $n \geq 2$.

In the pseudo-Jacobi case the conjectured parameter range is strictly wider than the range of the corresponding result in [2]: Conjecture 2 allows every a satisfying (8), whereas [2, Corollary 6.1] assumes $a < -n - 2$. The residual band

$$-n - 2 < a < \frac{-3n - 5 - \sqrt{n^2 + 2n + 5}}{4}$$

is nonempty for every $n \geq 1$, since $n+3 > \sqrt{n^2 + 2n + 5}$, and it is exactly the band on which the labels E_1, E_2 of [2] occur in reverse order; see Remark 4.3.

2. AN EXTERIOR-POINT CRITERION

Lemma 2.1 (Exterior-point criterion). *Let P, Q, G be monic real polynomials of degrees $n, n, n+1$, respectively, with $G \prec Q$. Suppose*

$$(1) \quad A(x)P(x) = -(x - c)G(x) + (x - r_-)(x - r_+)Q(x),$$

where $A(x) > 0$ for every $x \in \mathbb{R}$ and $r_- < r_+$. Assume that r_- and r_+ lie in distinct interior gaps of the zeros of G and that

$$(c - r_-)(c - r_+) > 0.$$

Then

$$\begin{aligned} c < r_- &\implies (x - r_-)G(x) \prec (x - r_+)P(x), \\ c > r_+ &\implies (x - r_+)G(x) \prec (x - r_-)P(x). \end{aligned}$$

The positivity hypothesis on A is less general than it appears. Since G, Q, P are monic, the two degree- $(n+2)$ terms on the right of (1) cancel, so $\deg AP \leq n+1$; a polynomial A is therefore of degree at most one, and positivity on $\mathbb{R}$ forces A to be a positive constant. This is what both applications below supply.

Proof. Let $y_1 < \dots < y_{n+1}$ be the zeros of G , put $R(x) = (x - r_-)(x - r_+)$, and set $F = RP$. Since $r_{\pm}$ lie in gaps of G and $c \notin [r_-, r_+]$, evaluating (1) at $r_{\pm}$ shows that $P(r_{\pm}) \neq 0$. At each y_j we have

$$(2) \quad A(y_j)F(y_j) = R(y_j)^2Q(y_j) \neq 0.$$

Because $G \prec Q$, the values $Q(y_j)$ alternate in sign. Hence F has a zero in every interval (y_j, y_{j+1}) .

Suppose first that $c < r_-$, and let $r_- \in (y_k, y_{k+1})$. The gap containing r_+ lies to the right, so

$$R(y_k) > 0 > R(y_{k+1}).$$

Monicity and $G \prec Q$ give

$$\operatorname{sgn} Q(y_k) = \operatorname{sgn} G(r_-) = -\operatorname{sgn} Q(y_{k+1}).$$

Using (1) at y_k, y_{k+1} and at r_- therefore yields

$$\operatorname{sgn} P(y_k) = \operatorname{sgn} P(y_{k+1}) = \operatorname{sgn} G(r_-), \quad \operatorname{sgn} P(r_-) = -\operatorname{sgn} G(r_-).$$

Thus P has a zero in each of (y_k, r_-) and (r_-, y_{k+1}) . Consequently, F has at least three distinct zeros in this gap and at least one in each of the other $n-1$ interior gaps. Since $\deg F = n+2$, these are all its zeros and all are simple. In particular, the gap containing r_+ has r_+ as its unique zero of F . Removing the zero r_- from $F = (x-r_-)(x-r_+)P$ shows that the zeros of $(x-r_+)P$ occupy exactly the intervals between consecutive zeros of $(x-r_-)G$. This proves the first conclusion. The case $c > r_+$ is symmetric. $\square$

3. MEIXNER–POLLACZEK POLYNOMIALS

Let $P_m^{(\lambda)}(x; \phi)$ be the monic Meixner–Pollaczek polynomial, normalised by $P_{-1}^{(\lambda)} = 0$, $P_0^{(\lambda)} = 1$ and the three-term recurrence

$$(3) \quad P_{m+1}^{(\lambda)}(x; \phi) = (x + (m+1)\cot \phi)P_m^{(\lambda)}(x; \phi) - \frac{m(m+2\lambda-1)}{4\sin^2 \phi} P_{m-1}^{(\lambda)}(x; \phi),$$

valid for $\lambda > 0$ and $\phi \in (0, \pi)$. This fixes the normalisation used below; for the family and its orthogonality weight see [1, §18.19]. For $n \geq 2$ and $\lambda > 0$, set

$$\phi_c = \frac{1}{2} \arccos\left(\frac{2\lambda - n - 1}{2\lambda + n + 1}\right) \in \left(0, \frac{\pi}{2}\right)$$

and assume

$$\phi \in (0, \phi_c) \cup (\pi - \phi_c, \pi).$$

The quadratic

$$(4) \quad T(x) = (1 - \cos 2\phi)x^2 + (n+1)\sin 2\phi x + \lambda^2 - (\lambda^2 + (n+1)\lambda)\cos 2\phi$$

has two distinct real roots $E_1 < E_2$ by [2, Lemma 5.1]. Here T is [2, (5.3)] and the displayed range of ϕ is [2, (5.4)], while the explicit roots [2, (5.7)–(5.8)] are labelled there with $E_1 < E_2$, so the two labellings agree. Conjecture 1 of [2] reads as follows, quoted in the source's notation.

Let $\{y_{k,n+1}\}_{k=1}^{n+1}$ denote the zeros of $P_{n+1}^{(\lambda+1)}(x; \phi)$. Let E_1, E_2 be as defined in (5.7) and (5.8), respectively, for $\lambda > 0$ and ϕ satisfying (5.4). If there exist distinct $k, k' \in \{1, \dots, n\}$ such that $y_{k,n+1} < E_1 < y_{k+1,n+1}$ and $y_{k',n+1} < E_2 < y_{k'+1,n+1}$, then

$$(x - E_1)P_{n+1}^{(\lambda+1)}(x; \phi) \prec (x - E_2)P_n^{(\lambda)}(x; \phi)$$

$$\text{or } (x - E_2)P_{n+1}^{(\lambda+1)}(x; \phi) \prec (x - E_1)P_n^{(\lambda)}(x; \phi)$$

holds.

Its hypothesis says that E_1 and E_2 lie in distinct interior gaps of the zeros of $P_{n+1}^{(\lambda+1)}(x; \phi)$, and its conclusion is the disjunction of the two alternatives below.

Theorem 3.1. *Suppose E_1 and E_2 lie in distinct interior gaps of the zeros of $P_{n+1}^{(\lambda+1)}(x; \phi)$. Then*

$$\begin{aligned} 0 < \phi < \phi_c &\implies (x - E_2)P_{n+1}^{(\lambda+1)}(x; \phi) \prec (x - E_1)P_n^{(\lambda)}(x; \phi), \\ \pi - \phi_c < \phi < \pi &\implies (x - E_1)P_{n+1}^{(\lambda+1)}(x; \phi) \prec (x - E_2)P_n^{(\lambda)}(x; \phi). \end{aligned}$$

In particular, Conjecture 1 of [2] holds.

Proof. Write

$$P = P_n^{(\lambda)}(x; \phi), \quad G = P_{n+1}^{(\lambda+1)}(x; \phi), \quad Q = P_n^{(\lambda+1)}(x; \phi), \quad c = \lambda \cot \phi.$$

The mixed recurrence in [2, (5.10)], together with $1 - \cos 2\phi = 2 \sin^2 \phi$, is

$$(5) \quad \frac{(2\lambda + n)(2\lambda + n + 1)}{2(1 - \cos 2\phi)} P = -(x - c)G + (x - E_1)(x - E_2)Q.$$

Both sides of (5) have degree at most $n + 1$, the degree- $(n + 2)$ terms on the right cancelling as noted after Lemma 2.1; the identity may therefore be checked from (3) at $n + 2$ points.

The left coefficient is positive. Also G and Q are Meixner–Pollaczek polynomials of consecutive degrees with the same parameters, hence orthogonal with respect to a common weight on $\mathbb{R}$ [1, §18.19], and the zeros of consecutive orthogonal polynomials separate each other [1, §18.2(vi)]; thus $G \prec Q$. Since $T(x) = (1 - \cos 2\phi)(x - E_1)(x - E_2)$, direct substitution gives

$$(c - E_1)(c - E_2) = \frac{T(c)}{1 - \cos 2\phi} = \frac{\lambda(2\lambda + n + 1)}{1 - \cos 2\phi} > 0.$$

Thus Lemma 2.1 applies. Moreover,

$$T'(c) = (2\lambda + n + 1) \sin 2\phi.$$

Because T opens upward and $c \notin [E_1, E_2]$, the derivative is positive exactly when $c > E_2$ and negative exactly when $c < E_1$. It is positive on $(0, \phi_c)$ and negative on $(\pi - \phi_c, \pi)$, so the two conclusions follow. $\square$

4. PSEUDO-JACOBI POLYNOMIALS

Let $P_m(x; \alpha, b)$ denote the monic pseudo-Jacobi polynomial; see [3, 2]. As in the previous section we fix the normalisation by a recurrence: throughout, $P_m(x; \alpha, b)$ is determined by $P_{-1} = 0$, $P_0 = 1$ and (6) below. We first record a proof of the consecutive-degree interlacing needed later, using nothing beyond that recurrence.

Lemma 4.1. *Let*

$$n \geq 1, \quad b \in \mathbb{R}, \quad \alpha < -n - \frac{1}{2}, \quad \alpha \neq -n - 1.$$

Then

$$P_{n+1}(x; \alpha, b) \prec P_n(x; \alpha, b).$$

Proof. We first record that (6) is the recurrence of the classical family, so that the normalisation fixed above is the usual one; the interlacing argument itself uses (6)–(7) alone. Comparing the pseudo-Jacobi definition with the Jacobi hypergeometric formula [1, (18.5.7)] gives

$$P_m(x; \alpha, b) = i^{-m} \hat{P}_m^{(\alpha+ib, \alpha-ib)}(ix),$$

where $\hat{P}_m^{(\mu, \nu)}$ is the monic Jacobi polynomial. The monic Jacobi recurrence coefficients [1, (3.5.33_1)] are rational in the parameters and therefore extend wherever their denominators are nonzero. Substitution yields, for $0 \leq m \leq n$,

(6)

$$P_{m+1}(x; \alpha, b) = \left(x + \frac{\alpha b}{(m + \alpha)(m + \alpha + 1)} \right) P_m(x; \alpha, b) - \gamma_m P_{m-1}(x; \alpha, b),$$

where $P_{-1} = 0$, $P_0 = 1$, and

$$(7) \quad \gamma_m = -\frac{m(m + 2\alpha)((m + \alpha)^2 + b^2)}{(m + \alpha)^2(2m + 2\alpha - 1)(2m + 2\alpha + 1)}.$$

The stated assumptions make every denominator in (6)–(7) nonzero. For $1 \leq m \leq n$, the three factors $m + 2\alpha$, $2m + 2\alpha - 1$, and $2m + 2\alpha + 1$ are negative, and hence $\gamma_m > 0$.

Thus P_m is the characteristic polynomial of the leading $m \times m$ principal block of a real symmetric tridiagonal matrix whose off-diagonal entries are $\sqrt{\gamma_m}$. The matrix is irreducible, so strict principal-minor interlacing gives the result. Equivalently, equality at a putative common zero would propagate backward through (6) to $P_0 = 0$, a contradiction. $\square$

Assume henceforth that

$$(8) \quad a < \frac{-3n - 5 - \sqrt{n^2 + 2n + 5}}{4}, \quad a \neq -n - 2,$$

and

$$(9) \quad |b| > \sqrt{\frac{4(a + n + 1)^2(a + n + 2)^2(-2a - n - 2)}{(n + 1)|4(a + n + 1)^2 - (2n - 2)(a + n + 1) - (n + 1)|}}.$$

Define

(10)

$$R_{n,a,b}(x) = x^2 - \frac{b(n + 1)}{(a + n + 1)(a + n + 2)}x + \frac{(a + n + 1)(a + n + 2)(2a + n + 2) + b^2(n + 1)}{(a + n + 1)(a + n + 2)(2a + 2n + 3)}.$$

Under (8)–(9), this polynomial has two distinct real roots by [2, Lemma 6.1]; write them in increasing order as $e_- < e_+$. Here (10), (8) and (9) are [2, (6.2), (6.3), (6.4)], and the explicit roots [2, (6.8)–(6.9)] are labelled there E_1, E_2 . Conjecture 2 of [2] reads as follows, quoted in the source's notation.

Let $\{y_{k,n+1}\}_{k=1}^{n+1}$ be the zeros of $P_{n+1}(x; a + 1, b)$. Let E_1 and E_2 be given by (6.8) and (6.9), where a and b satisfy the corresponding conditions (6.3) and (6.4). If there exist

distinct $k, k' \in \{1, \dots, n\}$ such that $y_{k,n+1} < E_1 < y_{k+1,n+1}$
and $y_{k',n+1} < E_2 < y_{k'+1,n+1}$, then

$$(x - E_1)P_{n+1}(x; a+1, b) \prec (x - E_2)P_n(x; a, b) \\ \text{or } (x - E_2)P_{n+1}(x; a+1, b) \prec (x - E_1)P_n(x; a, b)$$

holds.

Again the hypothesis says that the two completion points lie in distinct interior gaps of the zeros of $P_{n+1}(x; a+1, b)$, and the conclusion is a disjunction; see Remark 4.3 for the labelling.

Theorem 4.2. *Let $n \geq 2$, and suppose e_- and e_+ lie in distinct interior gaps of the zeros of $P_{n+1}(x; a+1, b)$. Set*

$$\Delta_c = \frac{b(2a + n + 3)}{(a + n + 1)(a + n + 2)}.$$

Then $\Delta_c \neq 0$ and

$$\Delta_c < 0 \implies (x - e_-)P_{n+1}(x; a+1, b) \prec (x - e_+)P_n(x; a, b), \\ \Delta_c > 0 \implies (x - e_+)P_{n+1}(x; a+1, b) \prec (x - e_-)P_n(x; a, b).$$

In particular, Conjecture 2 of [2] holds.

Proof. Put

$$u = a + n + 1, \quad c = \frac{b}{u},$$

and

$$P = P_n(x; a, b), \quad G = P_{n+1}(x; a+1, b), \quad Q = P_n(x; a+1, b).$$

The mixed recurrence [2, (6.10)], valid whenever its coefficients are defined, is

$$(11) \quad AP = -(x - c)G + R_{n,a,b}(x)Q,$$

where

$$A = \frac{2(2u - n - 1)(2u - n)(u^2 + b^2)}{u(2u - 1)(2u)(2u + 1)}.$$

The bound (8) is strictly smaller than $-n - \frac{3}{2}$, so $u < -\frac{1}{2}$ and therefore $A > 0$. It also gives $a + 1 < -n - \frac{1}{2}$, while $a \neq -n - 2$ gives $a + 1 \neq -n - 1$. Lemma 4.1 therefore yields $G \prec Q$.

A direct calculation gives

$$(12) \quad R_{n,a,b}(c) = \frac{(2u - n)(u^2 + b^2)}{u^2(2u + 1)} > 0.$$

Since $R_{n,a,b}$ is monic with roots $e_- < e_+$, it follows that $c \notin [e_-, e_+]$, and Lemma 2.1 applies. Finally,

$$R'_{n,a,b}(c) = \Delta_c.$$

For an upward-opening quadratic at a point outside its root interval, $R'(c) < 0$ exactly when $c < e_-$ and $R'(c) > 0$ exactly when $c > e_+$. The stated alternatives follow, and $\Delta_c \neq 0$. $\square$

Remark 4.3 (Ordering of the completion points). The formulas labelled E_1, E_2 in [2, (6.8)–(6.9)] are in increasing order when $a < -n - 2$ and in reverse order on the remaining admissible band. Using $e_- < e_+$ removes this notational reversal. Conjecture 2 is a disjunction of the two displayed interlacings and is invariant under exchanging the two labels, so Theorem 4.2 proves the conjecture exactly as stated.

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